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Chapter 1. 集合

$P(E) = \frac{\text{number of outcome for } E}{\text{total number in } \Omega} = \frac{K(E)}{n(\Omega)}$

$P(A\cap B) = P(AB) = \frac{K(AB)}{n(\Omega)}$ $P(A\cup B) = P(A) + P(B) - P(A\cap B)$ $P(A\cup B) = \frac{K(A\cup B)}{n(\Omega)}$

$P(\bar{A}) = 1 - P(A)$

$P(A|B) = \frac{P(A\cap B)}{P(B)}$ $(P(B)\neq 0)$：B条件下A的概率.

Chapter 4. 分布及标准误差:

$\Rightarrow$
- $\mu \pm \sigma \to 68.27\%$
- ~~$\mu \pm 2\sigma$~~ $\mu \pm 1.96\sigma \to 95\%$
- $\mu \pm 2.58\sigma \to 99\%$

$\mu \pm 3\sigma \to 99.7\%$

μ = mean value.
σ = SD.

SD (standard deviation) $\Rightarrow S = \sqrt{\frac{\Sigma(x_i-\bar{x})^2}{n-1}}$

$\Downarrow$ stand error $= \frac{SD}{\sqrt{n}}$

Sample Variance $= S^2 = \frac{\Sigma(x_i-\bar{x})^2}{n-1}$

CV (变异系数) $= \frac{SD}{\bar{x}} \cdot 100\%$
↳ 表示相对离散程度.

Z distribution $= \frac{x-\mu}{\frac{\sigma}{\sqrt{n}}}$

chapter. 6.：t分布 + 置信区间:

t-distribution $\Rightarrow t = \frac{\bar{x}-\mu}{\frac{S}{\sqrt{n}}}$

显著性水平 $\alpha = 0.05$ (假设 confidence $= 1-\alpha$ ex. $1-\alpha = 90\%$, $\alpha = 0.1$)

$\Downarrow$

Interval Estimation $\Rightarrow \left(\bar{x} - t_{\frac{\alpha}{2},v} \times \frac{S}{\sqrt{n}},\ \bar{x} + t_{\frac{\alpha}{2},v} \times \frac{S}{\sqrt{n}}\right)$

$v = n-1$ degree of freedom.

$t_{\frac{0.05}{2}}, v$

$H_0: \mu = 0$
$H_1: \mu \neq 0$. (there are any difference)

例除2. 一般都双侧 $\Leftarrow t_{0.05}, v$
- 问higher $\{\mu = \mu_1, \mu > \mu_1\}$
- 问lower $\{\mu = \mu_1, \mu < \mu_1\}$

Chapter 7.
t-test (t检验)

：解题步骤 - 5 steps.：

1. set up null and alternative hypothesis
   - null $\Rightarrow H_0$
   - alternative hypothesis $\Rightarrow H_1$

   the null hypothesis states that there is no difference of effect, while the alternative hypothesis represents the presence of a statistically significantly difference
2. Choose the level of significance. α $(\alpha = 0.05)$
3. calculate the test statistic
   ↳ $t = \frac{\bar{x}-\mu_0}{\frac{S}{\sqrt{n}}}$
4. 查critical value. $\Rightarrow t_{0.025}, df = t_{0.025}, n-1$；t表入.
5. 结论. $\Rightarrow$
   - $|t| > t_{\frac{\alpha}{2},df} \Rightarrow$ Reject H_0
   - $|t| \leq t_{\frac{\alpha}{2},df} \Rightarrow$ Fail Reject H_0.

one-sample t-test

two-sample t-test.

paired-sample t-test
$\Rightarrow t = \frac{\bar{d}}{S_d/\sqrt{n}}$, $\bar{d} = \frac{\Sigma d_i}{n}$, $d_i = x_{1i} - x_{2i}$

Independent t-test $\Rightarrow t = \frac{\bar{x}_1 - \bar{x}_2}{S_{\bar{x}_1-\bar{x}_2}}$ $S_{\bar{x}_1-\bar{x}_2} = \sqrt{S_c^2 \times \left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$ $S_c^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$

$S_{1或2} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$

P(type I error) = significance level $= \alpha = 0.05$. $\Rightarrow$ 错误拒绝正确的零假设.

P(type II error) $= \beta \Rightarrow$ 未能拒绝错误的零假设.

Chapter 8. ANOVA： step1. 提出假设.
- $H_0: \mu_1 = \mu_2 = \mu_3 \cdots \mu_n$
- H_1：至少存在一组的均数与其它组不同.

检验三组及以上的组间均值差异.

$MS_b = \frac{SSB}{k-1}$ $SSB = \sum_{i=1}^{k} n_i(\bar{x}_i-\bar{x})^2$ step2. chouse $\alpha = 0.05$.

step3. calculate F value $\to F = \frac{MS_{bet}}{MS_w}$
- MS_{bet} → 组间方差
- MS_w → 组内方差

$df_{bet} = k - 1$ → k 组数.

$df_w = N - k$ → N 总样本量.

查表得出 t 临界值

$MS_w = \frac{SSW}{N-k}$

$SSW = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij}-\bar{x}_i)^2$

Step4：F与 t 临界值对比.

Step5：F > t 拒绝 H_0，支持 H_1
F < t 未能拒绝 H_0

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Chapter. 9. Chi-square Test：解题步骤：

1. set the hypothesis $H_0: \pi_T = \pi_C$ $H_1: \pi$…
2. set $\alpha = 0.05$
3. Calculate χ^2 & df → chi-square fit：$df = k$… ；列联表：$df = (r-1)\cdot(c$…$)$ （r → 行，c → 列）
4. 查表（根据 df & 0.05）→ P194
5. 对比＋总结.

χ^2 值 > 查表的 critical value 便可拒绝 H.

$$\chi^2 = \sum \frac{(f_O - f_E)^2}{f_E}$$

f_O → 实际发生　　f_E → 理论期望.

连续性校正 χ^2 公式：$\chi^2_{corred} = \sum \frac{(|f_O - f_E| - \frac{n}{2})^2}{f_E}$

① $n \geq 40$ / $f_E \geq 5$ 不需要 $\chi^2_{correct}$

② $n \geq 40$ / f_E 有个 <5 需要 $\chi^2_{correct}$
$1 < f_E < 5$

③ ~~$n < 40$ or 1个 $f_E < 1$ 需要 fisher's exact~~. → 计算复杂，一定不会考这个。

列联表 → 每个格子算期望频率.

a	b	a+b
c	d	c+d
a+c	b+d	

$f_{Ea} = \frac{(a+c)\cdot(a+b)}{a+b+c+d}$

~~$f_a = a$~~

Chapter. 10. Nonparametric statistic

① Wilcoxen signed Rank Test

1. setup H_0 / H_1 & $\alpha = 0.05$
2. calculate the difference
3. Rank
4. 正差值 → 正秩　负差值 → 负秩
5. 取较小者 计算：T^+ & T^-、查表.

② Mann-whitney U test

① 合并数据后再排序，时秩

② 算秩和 — R_1（一组秩和）；R_2（二组秩和）

列 → n_1, n_2

$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$ ⇒ 第1组秩和.

$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$

Chapter 11. Correlation & simple Linear Regression.

① A Linear correlation

正相性
负相性　⇒ $r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i-\bar{x})^2 \sum(y_i-\bar{y})^2}}$
无线性相关

ex 弱 0 - 0.19
　 弱 0.20 - 0.39
　 中 0.40 - 0.59
　 强 0.6 - 0.79
ex 强 0.8 - 1

② Rank correlation. ⇒ $r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$

Simple linear regression:

$Y = \alpha + \beta x + \epsilon$ ⇒ 误差项.

α → population intercept ‖ 截距　　β → population regression coefficient.

⇒ 估计公式 ⇒ $\hat{y} = b_0 + b_1 x$

$b_1 = \frac{SS_{xy}}{SS_{xx}}$　SS_{xy} → $\sum(\bar{x}-x_i)(\bar{y}-y_i)$；$SS_{xx}$ → $\sum(\bar{x}-x_i)^2$

$b_0 = \bar{y} - b_1\bar{x}$

Hypothesis Testing on the regression coefficient.

1. set up the hypothesis significant level! ⇒ $H_0: \beta = 0$ $H_1: \beta \neq 0$ $\alpha = 0.05$.
2. Calculate the testing statistic t: $t = \frac{\beta_1 - 0}{SE(\beta_1)}$ → $\sqrt{\frac{SSE}{(n-2)\sum(x_i - \bar{x})^2}}$
3. Review t 以临界值 & make decision.
4. conclusion.

$SSE = \sum(y_i - \hat{y}_i)^2$

…y Error: Sampling error is the difference between a sample statistic (such as the sample mean or proportion) and …e population parameter that arises because the sample does not include every member of the population. …curs due to random variation in the selection of the sample and decreases as the sample size increases.

…factors that determine the sampling error: Sampling error is mainly determined by the sample size, the variability the ~~poplu~~ population, and the sampling method. A large sample size generally leads to a smaller sampling error.

Type. I error: A Type I error occurs when the null hypothesis is incorrectly rejected even though it is true. It is also called a false positive. The probability of making a Type I error is denoted by the significance level α.

Type II error: A Type II error occurs when the null hypothesis is not rejected even though it is false. The type II error is denoted by β.

deviation: 某个数据点与Mean value的差。

Variety of dispersion: describes how spread out the data values are around a measure of central tendency.

The coefficient of variation: is a dimensionless measure of relative dispersion, defined as the ratio of the standard deviation to the mean.

$\rightarrow CV = \frac{SD}{\bar{x}} \cdot 100\%$. $\Rightarrow SD = \sqrt{\frac{\Sigma(x_i-\bar{x})^2}{n-1}}$

Standard deviation measures the average spread of data values around the mean.

Sample variance is the average of the squared deviations from the sample mean. $\rightarrow s^2 = \frac{\Sigma(x_i-\bar{x})^2}{n-1}$

Mean deviation: is the average of the absolute deviations of observations from a central value. $M = \frac{\Sigma|x_i-\bar{x}|}{n}$.

The Z distribution is the standard normal distribution with mean 0 and standard deviation 1, obtained by standardising a normal variable.

Categorical variable is a variable that takes on a limited number of distinct categories or labes.

The Central limit theorem states that the sampling distribution of the sample mean approaches a normal distribution as sample size increase.

Standard error: measures the variability of a sample statistic (eg. mean) as an estimate of the population parameter.

A confidence interval is a range of values derived from sample data that is likely to contain the true population parameter with specified probability.

A critical value is the threshold that a test statistic must exceed to reject the null hypothesis at a specific significance level.

Interval estimation is a method of estimating a population parameter by providing a range of value that is likly contain the true parameter.

The normal distribution: is a continuous, symmetric, bell-shaped probability distribution defined by. It's mean and SD

parameter estimation is the process of using sample data to estimate unknow population parameter.

A hypothesis test: is a statistical procedure used to decide whether to reject a null hypothesis based on sample data.

p-value is the probability of obtaining the observed result, or one more extreme, assuming the null hypothesis is true.

A scattergraph is a graphical display that shows the relationship between quantitative variable using plotted points.

The linear correlation coefficient measures the strength and direction of the linear relationship between two quantitative variables.

Chi-squre test is a statistical method used to access associations between categorical variables or the fit ~~between cate~~ between observed and expected frequencies.

④ A clinical trial investigates to ---- LDL Cholesterol (mg/dL) { control (n=10): Mean=135 SD=17. Diet (n=12): Mean=121. SD=15

----- $\alpha=0.05$ provide pooled variance, t-statistic, df.

A: ① $H_0: \mu_1=\mu_2$
$H_1: \mu_1\neq\mu_2$
$\alpha=0.05$

② $S_p^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2} = 188.55$ → $S = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$

③ $\nu = n_1+n_2-2 = 20$.

④ $t = \frac{\bar{x}_1-\bar{x}_2}{\sqrt{S_p^2(\frac{1}{n_1}+\frac{1}{n_2})}} = \frac{14}{\sqrt{188.55\cdot 0.183}} \approx 2.381$

⑤ $t_{\frac{0.05}{2},20} = 2.086$
$\Rightarrow |t| \approx 2.381 > 2.086$

⑥ 拒绝H_0.

⑤ From previous studies, it has been determined that the level of glucose in the blood has a mean $\mu_0 = 9.7$ mmol/L. ---- Sample of n=25 ---- Sample mean = 13.6 mmol/L and SD = 2 mmol/L, is it likely that came from population with a mean 9.7 or not

A: ① $\mu_0 = 9.7$ mmol/L
$n=25$
$\bar{x} = 13.6$ mmol/L
$SD = 2$ mmol/L
$\nu = n-1 = 25-1 = 24$.

② $H_0: \mu = 9.7$
$H_1: \mu \neq 9.7$
$\alpha = 0.05$

③ $t = \frac{\bar{x}-\mu_0}{S/\sqrt{n}}$

$t = \frac{13.6-9.7}{2/\sqrt{25}} = \frac{3.9}{\frac{2}{5}} = \frac{3.9}{0.4} = 9.75$

④ $t_{0.05/2,24} = 2.064$.
$\because |t| = 9.75 > 2.064$
$\therefore$ 拒绝H_0.

⑥ 配对用差值 $\bar{d} = 6+9+11+5+9+7+7+9+10+7 = 76 \div 10 = 7.6$.

$S_d \approx 1.776$.

$t = \frac{\bar{d}}{S_d/\sqrt{n}} = \frac{7.6}{1.776/3.162} = \frac{7.6}{0.562} \approx 13.52$

$\nu = 10-1 = 9$ $t_{0.05,9} = 1.833$

13.52
~~$t \approx 13.52$~~ > 1.833
re H_0

① According to large-scale survey data, it is know that the average birth weight of general infants is $\mu_0 = 3.3$ kg ---- 36 dystocia infants ----- Sample mean 3.43 kg, SD 0.4 kg. Is there any difference ----?

A: ∵ know $\mu_0 = 3.3$ kg
$n=36$
$\bar{x} = 3.43$ kg
$SD = 0.4$ kg
$\nu = 36-1 = 35$

① $(H_0) = \mu = \mu_0$
$(H_1) = \mu \neq \mu_0$
$\alpha = 0.05$

② $t = \frac{\bar{x}-\mu_0}{\frac{S}{\sqrt{n}}} = \frac{3.43-3.3}{\frac{0.4}{\sqrt{36}}} = \frac{0.13}{0.0667} \approx 1.95$

③ $t_{\frac{0.05}{2},35} \approx 2.03$
$|t| = 1.95$

④ $|t| < t_{\frac{0.05}{2},35}$ ∴ $\alpha=0.05$ 时 不 reject H_0 ∴不能认为难产儿的出生体重与一般有差异。

② From previous studies, it has been determined that ----- mean $\mu_0 = 7$ mmol/L --- Sample of n=25 --- sample mean = 9.6 mmol/L ---- SD = 2 mmol/L, is it likely that the level of glucose in diabetic patients is higher than then general population.

① $H_0 = \mu = 7$ mmol/L
$H_1 = \mu > 7$ mmol/L
$\alpha = 0.05$

② $t = \frac{\bar{x}-\mu}{\frac{S}{\sqrt{n}}} = \frac{9.6-7}{2/\sqrt{25}} = 6.5$

③ $\nu = 24$ $\alpha = 0.05$
$t_{0.05,24} = 1.711$
$t = 1.711$

$t = 6.5 > 1.711$
∴ 拒绝H_0, 接受H_1

③ two sample of 12-year-old boys were drawn in one city in 1973 & 1983, Compare the height ~~information~~ difference between the two sample.

{ 1973: $n_1 = 120$ $\bar{x}_1 = 139.9$ cm $S_1 = 7.5$ cm
1983: $n_2 = 153$ $\bar{x}_2 = 143.7$ cm $S_2 = 6.3$ cm.

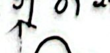
① $H_0: \mu_1 = \mu_2$
$H_1: \mu_1 \neq \mu_2$
$\alpha = 0.05$

② $\nu = n_1+n_2-2 = 271$
$t_{\frac{0.05}{2},271} \approx 1.96$

③ $|t| = \frac{\bar{x}_1-\bar{x}_2}{\sqrt{\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}}} = \frac{139.9-143.7}{\sqrt{\frac{7.5^2}{120}+\frac{6.3^2}{153}}} = \frac{-3.8}{\sqrt{0.46}} \approx 4.45$

④ ∵ $|t| = 4.45 > 1.96$
∴ 拒绝H_0, 接受H_1,
存在显著差异。

⑥ Suppose that IQ Scores in a certain population are Normally distributed with mean 100 and SD=15, why might a scientist not refer to an IQ of 89 as low-IQ (~~$P_z < -0.73$~~) $P_{z<0.73} = 0.2327$). $P_{z<-0.73} = 0.2327 \approx 23.27\%$

① 画图 

② ∵ $\mu = 100$
$\sigma = 15$
$x = 89$

∵ $Z = \frac{x-\mu}{\sigma}$
$Z = \frac{89-100}{15} = -0.73$
∴ $Z \approx -0.73$

大约23%的人低于89.

...inus varible.

~~Mean for sample of sig~~

...escription of categoral varible.

$$Ratio = \frac{A\ index}{B\ index} \cdot (100\%) \quad proportion = \frac{one\ part}{whole\ part} \times 100$$

$$Rate = \frac{Event\ occurs\ in\ period}{whole\ number\ of\ event\ to\ occur.}$$

$$MoRality\ Ratio = p' = p \frac{G}{Exp.} = p_e\ Sap.$$

Non-parametric test

$$U = \frac{T - \frac{1}{2} \cdot n_1(N+1)}{\sqrt{\frac{n_1 n_2 (N+1)}{12}} \left(1 - \frac{\Sigma(t_j^3 - t_j)}{N^3 - N}\right)}$$

T-test diff b/w two group.

$$Z = \frac{\bar{x} - \mu_0}{G/\sqrt{n}} \sim N(0,1)$$

$$t = \frac{x^2 - \mu_0}{G/\sqrt{n}} \sim t(0,1).$$

One sample T test.

$$Z = \frac{\bar{x} - \mu_0}{G/\sqrt{n}} \rightarrow G\ known$$

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$$

$v = n-1.$

Two sample T test.

Dependant: (paired Sample t-test)

$H_0: \mu_1 = 0$

$H_1: \mu_2 \neq 0$

$$t = \frac{\bar{d}}{Sd/\sqrt{n}}$$

$\bar{d} = x_i - y_i$

$$\bar{d} = \frac{1}{n}\left(\sum_{i=1}^{n} d_i - n\bar{d}^2\right)$$

$t_{\frac{\alpha}{2}}, t_{\alpha}, -t_{\alpha}.$

In dependendant.

$G_1^2 = G_2^2$

$H: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2.$

$$t = \frac{\bar{x}_1^2 - \bar{x}_2}{S_{x_1} - S_{x_2}} = \frac{\bar{x}_1 - \bar{x}_2}{S_2\sqrt{(\frac{1}{n_1} + \frac{1}{n_2})}}$$

$$S_{x_1} - S_{x_2} = \sqrt{SC(\frac{1}{n_1} + \frac{1}{n_2})}$$

$$SC^2 = \frac{(n_1 - 1)S^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$v = n_1 + n_2 - 2$

$$S_1^2 = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n-1}}$$

Chi-square Test.

$$\chi^2 = \frac{\Sigma(O - E)^2}{E} \quad (n \geq 40 / T \geq 5)$$

$$\chi^2 = \frac{\Sigma(|A - T| - 0.5)^2}{T} \quad n \geq 40 \quad 1 \leq T < 5.$$

$V = (R-1)(C-1)$

$$T_{RC} = \frac{n_R n_C}{n}$$

Special formula.

$$\chi^2 = \frac{(ad - bc)^2 n}{(a+b)(a+c)(b+d)(c+d)}$$

$$\chi_c^2 = \frac{(|ad - bc| - \frac{n}{2})^2 n}{(a+b)(a+c)(b+d)(c+d)}$$

For paired sample.

2×2 Nemn's test.

(1) $\frac{a+b}{a+b+c+d}$ $\frac{a+c}{a+b+c+d}$.

$$\chi^2 = \frac{(b-c)^2}{b+c} \quad v = 1$$

$$\chi_c^2 = \frac{(|b-c| - 1)^2}{b+c} \quad v = 1.$$
